

Definition and first terms

$9\,A(x) = 8 + 8\,x + A(x)^5$, with the origin-normalized solution.
 Normalization/grammar: $A(x)=1+(2)*T(x)$; $T=x+5*T^2+10*T^3+10*T^4+4*T^5$.
 $a(0),\dots,a(7) = 1, 2, 10, 120, 1770, 29208, 516180, 9554640$.
 Kernel used throughout: $\rho(u) = u\,(1-5\,u^1-10\,u^2-10\,u^3-4\,u^4)$, so $x = \rho(T(x))$.

Typogeometry

1x2

{1,1}x10

{1,{1,1}}x50

{1,1,1}x20

colored arity geometry; node color distinguishes constructors. Filled endpoints are true leaves; open endpoints are false/empty leaves. For colored grammars, color is genuine constructor data and is not silently converted into a positional zero pattern.

Typogeometric constructor codes and multiplicities: Δ_2 : 5, Δ_3 : 10, Δ_4 : 10, Δ_5 : 4.

Coefficient and contour extraction

$$\mathcal{K}_n = \{(k_1, \dots, k_4) \in \mathbb{Z}_{\geq 0}^4 : \sum_{j=1}^4 j k_j = n - 1\}, \quad K = \sum_{j=1}^4 k_j,$$

$$a(n) = \frac{2}{n} \sum_{\mathbf{k} \in \mathcal{K}_n} \binom{n + K - 1}{K} \binom{K}{k_1, \dots, k_4} 5^{k_1} 10^{k_2} 10^{k_3} 4^{k_4}.$$

$$a(n) = \frac{2}{2\pi i n} \oint_{\gamma} \frac{du}{\rho(u)^n}, \quad n \geq 1.$$

Recurrence

$$\sum_{r=0}^4 P_r(n) a(n+r) = 0, \, n \geq 1, \text{ with}$$

$$\begin{aligned} P_0(n) &= -50000\,n^4 - 200000\,n^3 - 220000\,n^2 - 40000\,n + 18480 \\ P_1(n) &= -200000\,n^4 - 1100000\,n^3 - 2140000\,n^2 - 1750000\,n - 510000 \\ P_2(n) &= -300000\,n^4 - 2100000\,n^3 - 5370000\,n^2 - 5910000\,n - 2340000 \\ P_3(n) &= -200000\,n^4 - 1700000\,n^3 - 5200000\,n^2 - 6700000\,n - 3000000 \\ P_4(n) &= 9049\,n^4 + 90490\,n^3 + 316715\,n^2 + 452450\,n + 217176 \end{aligned}$$

Differential equation

The scalar linear ODE has order 4. It is obtained from the recurrence by the standard Euler-operator substitution. Exact coefficients and boundary polynomial: `data/ode.json`.

Matrices, telescoper certificate, and checks

No compact matrix tuple is shown; see the embedded exact reduction data.

Certificate.

$R(n,u) = N(n,u)/\rho(u)^3$, $\deg_u N = 16$.
 Telescoping identity: $\sum_r P_r(n) H_{n+r}(u) = \partial_u (R(n,u) H_n(u))$.
 Matrix dimensions: 10×10 ; remainder 4×5 , rank 4, nullity 1. The exact telescoping residual is zero. The recurrence reconstructs all 24 stored terms; 6/6 published OEIS prefix terms match.